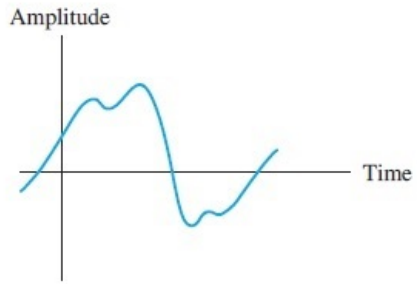
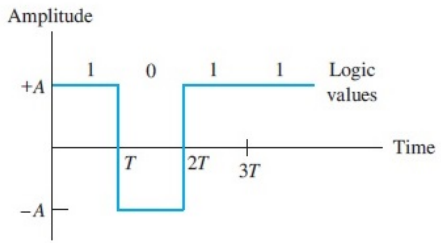
Introduction

- The amplitude of analog signal in a continuous range has a unique significance. For instance, a position sensor may produce an analog signal that is proportional to displacement, and each amplitude represents a different position.



Analog signal

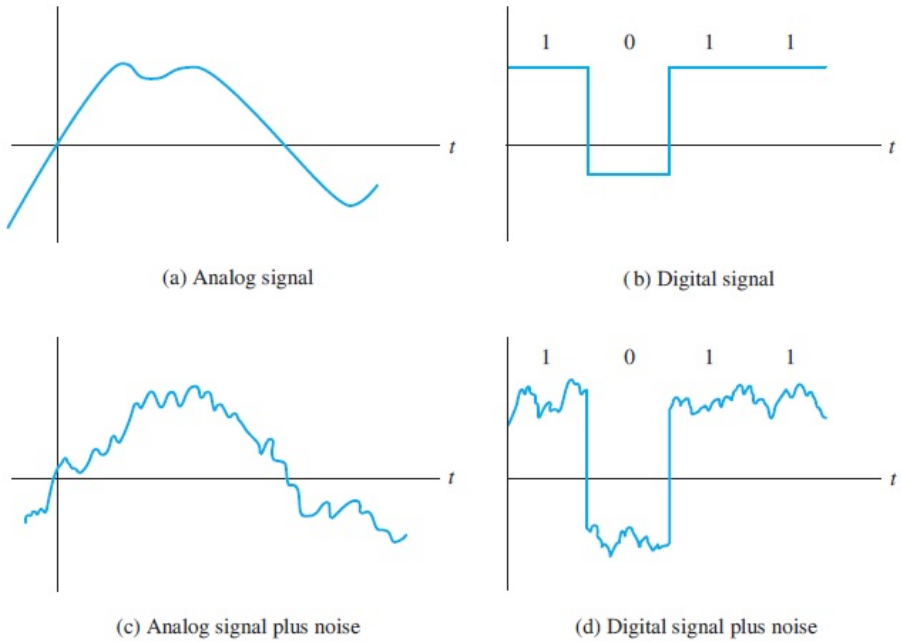
- For a digital signal, only a few restricted ranges of amplitude are allowed and each amplitude in a given range has the same significance. Most common are binary signals that take on amplitudes in only two ranges. The information associated with the ranges is represented by the logic values 1 or 0.



Digital signal

Advantages of the Digital Approach

- After noise is added to an analog signal, it is usually impossible to determine the precise amplitude of the original signal. However, after noise is added to a digital signal, it is still possible to determine the logic values so long as the noise amplitude is not too large.



(a) Analog signal (b) Digital signal

(c) Analog signal plus noise (d) Digital signal plus noise

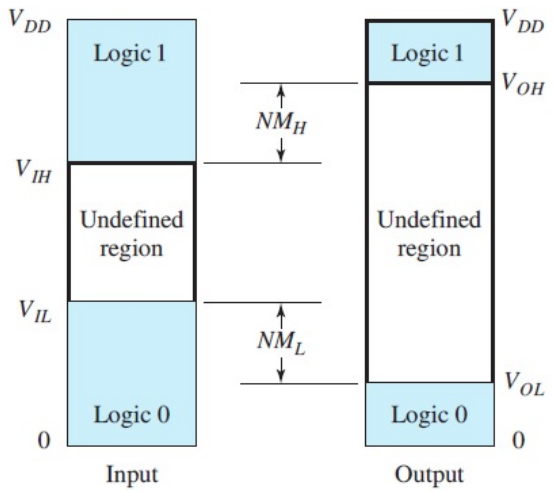
- For a given type of logic circuit, one range of voltages represents logic 0 and another range of voltages represents logic 1. For a proper operation, a logic circuit only needs to produce a voltage somewhere in the correct range. Therefore, component values in digital circuits do not need to be as precise as in analog circuits.

Advantages of the Digital Approach

- With modern integrated-circuit (IC) manufacturing technology, very complex digital logic circuits can be produced economically. However, analog circuits often needs large capacitances and precise component values that are impossible to manufacture as large scale integrated circuits. Therefore, digital systems have become increasingly important in the past few decades.
- The logic value 1 is also called HIGH, TRUE or ON. The logic 0 is also called LOW, FALSE or OFF. The signals in logic systems switch between high and low as the information being represented changes. These signals or 'logic variables' are denoted by uppercase letters such as A, B and C.

Logic Ranges and Noise Margins

- Logic circuits are typically designed so that a range of input voltages is accepted as logic 1 and another nonoverlapping range of voltages is accepted as logic 0. The largest input voltage accepted as logic 0 or LOW is denoted by V_{IL} and the smallest input voltage accepted as logic 1 or HIGH is denoted by V_{IH}. No meaning is assigned to voltages between V_{IL} and V_{IH}, which normally occur during transitions.



- Moreover, the digital circuits are designed such that the output voltages fall into narrower ranges than the inputs (Noise can be added to a logic signal in the interconnections between outputs and inputs, it is important that the outputs have narrower ranges than the acceptable inputs).

Combinatorial Logic Circuits

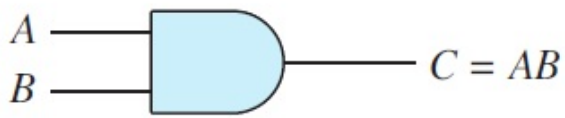
- In this section, we consider circuits called logic gates that combine several logic variable inputs to produce a logic-variable output.
- The circuits that we are about to discuss are said to be memoryless since their output values at a given instant depend only on the input values at that instant.

AND Gate

- The AND operation on two logic variables A and B is represented as AB, read as 'A and B'. The AND operation is also called logical multiplication.
- One way to specify a combinatorial logic system is to list all the possible combinations of the input variables and the corresponding output values. Such a listing is called a 'truth table'. The truth table of the AND gate and its circuit symbol is given below:

A	B	$C = AB$
0	0	0
0	1	0
1	0	0
1	1	1

(a) Truth table



(b) Symbol for two-input AND gate

AND Gate

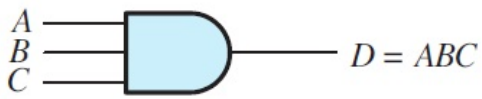
- For the AND operation, we can write the following equations:

$$AA = A$$
$$A1 = A$$
$$A0 = 0$$
$$AB = BA$$
$$A(BC) = (AB)C = ABC$$

- The truth table for a three-input AND gate is given below.

A	B	C	$D = ABC$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

(a) Truth table

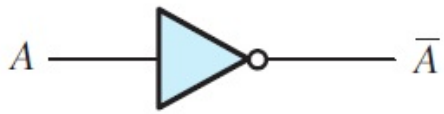


Logic Inverter

- The NOT operation on a logic variable is represented by placing a bar over the symbol for the logic variable. The symbol $\bar{A}$ is read as 'not A' or as 'A inverse'. If A is 0, $\bar{A}$ is 1, and vice versa. The truth table and the symbol of the inverter is given below.

A	$\bar{A}$
0	1
1	0

(a) Truth table



(b) Symbol for an inverter

- For the NOT operation, we can write the following equations:

$$A\bar{A} = 0$$

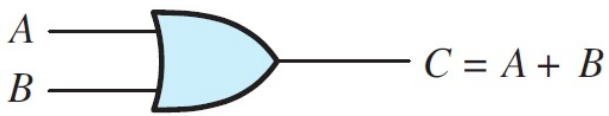
$$\bar{\bar{A}} = A$$

OR Gate

- The OR operation of logic variables is written as $A + B$, which is read as 'A or B'. The truth table and the circuit symbol of a two-input OR gate is shown in the figure below. Notice that $A + B$ is 1 if A or B (or both) are 1. The OR operation is also called logical addition.

A	B	$C = A + B$
0	0	0
0	1	1
1	0	1
1	1	1

(a) Truth table



(b) Symbol for two-input OR gate

OR Gate

- For the OR operation, we can write the following equations:

$$(A + B) + C = A + (B + C) = A + B + C$$

$$A(B + C) = AB + AC$$

$$A + 0 = A$$

$$A + 1 = 1$$

$$A + \bar{A} = 1$$

$$A + A = A$$

- Boolean algebra expressions can be implemented by interconnection of AND gates, OR gates, and inverters.
- Consider the logic expression
  $F = (AB + CA)\bar{B} + C(A + \bar{C}) + (A + C)(B + \bar{C})$.
- We can write down its truth table as follows:

A	B	C	$\bar{B}$	$\bar{C}$	AB	CA	$AB + CA$	$(AB + CA)\bar{B}$	$A + \bar{C}$	$C(A + \bar{C})$	$A + C$	$B + \bar{C}$	$(A + C)(B + \bar{C})$	F
0	0	0	1	1	0	0	0	0	1	0	0	1	0	0
0	0	1	1	0	0	0	0	0	0	0	1	0	0	0
0	1	0	0	1	0	0	0	0	1	0	0	1	0	0
0	1	1	0	0	0	0	0	0	0	0	1	1	1	1
1	0	0	1	1	0	0	0	0	1	0	1	1	1	1
1	0	1	1	0	0	1	1	1	1	1	1	0	0	1
1	1	0	0	1	1	0	1	0	1	0	1	1	1	1
1	1	1	0	0	1	1	1	0	1	1	1	1	1	1

Implementation of Boolean Expressions

- Note that the logical expression can be simplified as follows.

$$\begin{aligned} F &= (AB + CA)\bar{B} + C(A + \bar{C}) + (A + C)(B + \bar{C}) \\ &= AB\bar{B} + CA\bar{B} + CA + C\bar{C} + AB + A\bar{C} + CB + C\bar{C} \\ &= CA\bar{B} + CA + AB + A\bar{C} + CB \\ &= A(C + B + \bar{C}) + C(A\bar{B} + B) \\ &= A + C(A\bar{B} + B) \\ &= A + BC \end{aligned}$$

- It is easier to obtain the truth table of F from the simpler expression.
- Exercise: Obtain the truth table of F from the simpler expression.

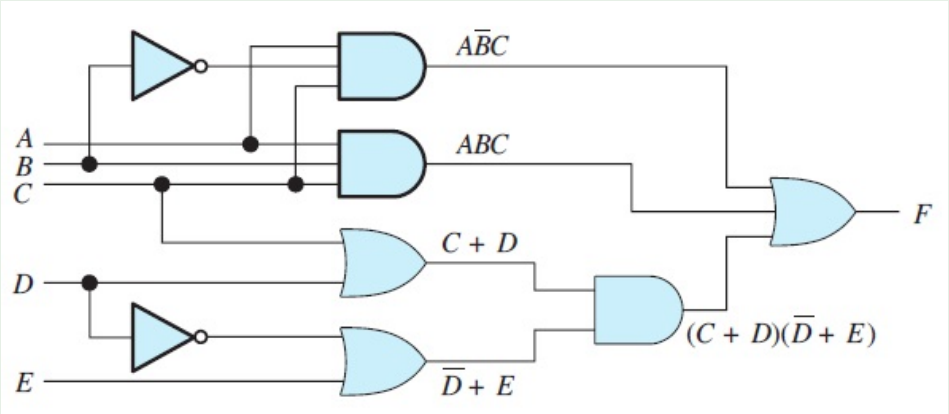
Implementation of Boolean Expressions

Example

Consider the logic expression

$$F = A\bar{B}C + ABC + (C + D)(\bar{D} + E)$$

It can be implemented by the logic circuit shown in the figure below:



However, sometimes it is possible to manipulate a logic expression to find an equivalent expression which is simpler. For instance, we can expand the logic expression of F as

$$F = A\bar{B}C + ABC + C\bar{D} + CE + D\bar{D} + DE$$

Implementation of Boolean Expressions

Since $D\bar{D} = 0$, we can drop it from the expression. Furthermore, we can factor the first two terms as follows

$$F = AC(\bar{B} + B) + C\bar{D} + CE + DE.$$

Since $\bar{B} + B = 1$, we can write

$$F = AC + C\bar{D} + CE + DE$$

If we factor C from the first three terms, we have

$$F = C(A + \bar{D} + E) + DE.$$

Note that it is often possible to find alternative implementations for a given logic function. There are methods for finding the implementation using the fewest number of gates of a given type.

Implementation of Boolean Expressions

Homework Exercise: Give the truth table for the boolean function F.

Answer:

A	C	D	E	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Implementation of Boolean Expressions

Example

Given the truth table below, a) determine the logic expression of F, b) implement the logic circuit by gates.

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Implementation of Boolean Expressions

Solution:

For a $F = 1$ value in the truth table, we express the combination of the logic values A, B and C by connecting them with AND gates. Then we connect these expressions by OR gates to derive the expression of F:

$$F = \bar{A}BC + A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C} + ABC$$

Implementation is assigned as a homework.